

Division properties of exponents and zero exponent

1 Simplify the following, giving your answers in exponential form.

a $10^8 \div 10^6$

b $j^8 \div j^4$

c $a^5 \div a^3$

d $9q^6 \div (4q^2)$

2 Simplify the following, giving your answers with positive exponents.

a $\frac{18^{32}}{18^{23}}$

b $\frac{a^{10}}{a^6}$

c $\frac{u^{13}}{u^2}$

d $\frac{t^7 r^8}{t^5 r^4}$

e $\frac{5r^7}{r^4}$

f $\frac{3j^{13}}{4j^8}$

g $\frac{-x^{11}}{3x^4}$

h $\frac{9j^9 k^8}{4j^7}$

i $\frac{3g^8}{2g^5 h^6}$

j $\frac{9j^5 k^7}{j^3 k^4}$

k $\frac{j^8 k^9}{7j^7 k^8}$

l $\frac{3m^9 n^4}{8m^8 n^2}$

m $\frac{4p^3 q^9}{2p^2 q^7}$

n $\frac{2p^9 q^6}{6p^3 q^3}$

o $\frac{3j^6 k^4}{2j^4 k^2}$

p $\frac{y^6}{(4y)^2}$

q $\frac{6r^3 s^6}{2r^2 s^4}$

r $\frac{-9x^{13}}{3x^4}$

s $\frac{x^6}{4x^4}$

t $\frac{54x^{11}}{6x^6}$

3 For the following, write the integer value or the term that should go in the space.

a $a^9 \div a^1 = a^7$

b $15j^{14} \div (\text{I}) = 5j^7$

c $x^{23} y^{10} \div (\text{II}) = x^{15} y^5$

d $\text{II} \div (g^4 h^3) = g^{16} h^5$

e $63x^{18} \div \text{III} = 9x^{14}$

f $x^{21} y^8 \div \text{III} = x^{14} y^3$

g $\frac{8r^3 s^8}{\text{IV}} = 2rs^2$

4 Simplify the following, giving your answers in exponential form:

a $\frac{(n^8 r^5)^5}{(n^4 r)^5}$

b $\frac{30x^{13} y^{26} z^{21}}{3x^{11} y^{15} z^6}$

c $\frac{15x^{16} y^{25} z^{13}}{-3x^{14} y^{11} z^2}$

d $(m^{12})^9 \div (m^4)^2$

e $(-240u^{32}) \div (-8u^9) \div (-5u^{12})$

f $\frac{y^7 \cdot y^5}{y^3 \cdot y^2}$

g $\frac{10p^6 \cdot 3p^{10}}{15p^2}$

h $\frac{(2x^4 y^0)^2}{x^6}$

i $\frac{6(w^2 v)^5 \cdot 27y^{17} v^7}{(3w^5 v^3)^2 \cdot 2y^6 v}$

j $m^9 \div m^5 \cdot m^4$

k $p^{18} \div p^8 \div p^5$

l $\frac{(x^3)^2}{x}$

m $\frac{45b^6}{(3b)(5a)}$



$\frac{210p^{22}}{7p^8} \div (5p^7)$

- 5 Fill in the missing value to complete the pattern.

$k^5 = 1 \cdot k \cdot k \cdot k \cdot k \cdot k$

$k^4 = 1 \cdot k \cdot k \cdot k \cdot k$

$k^3 = 1 \cdot k \cdot k \cdot k$

$k^2 = 1 \cdot k \cdot k$

$k^1 = 1 \cdot k$

$k^0 = 1$

- 6 Fill in the missing exponent to make the following equations true.

a $p^{12} \div p^{12} = p^1$

b $\frac{b^1}{b^{11}} = b^0$

c $\frac{w^7}{w^1} = w^0$

- 7 Evaluate:

a 741^0

b $(-983)^0$

c $(2 \cdot 13)^0$

- 8 Simplify:

a $18a^0$

b q^0

c $11a^0$

d $(6a)^0$

e $(a^0)^{79}$

f $9 \cdot (15x^6)^0$

g $(3m^4)^3 \cdot mq^0$

h $\frac{45x^7y^7}{9x^7y^2}$

Additional questions

- 9 When simplifying the expression $\frac{a^5 \cdot a^4}{a^2}$, Chad's first line of work was $a^3 \cdot a^2$. Is Chad's work correct? Explain.

- 10 When simplifying the expression $\frac{a^5 + a^4}{a^2}$, Clarice's first line of work was $a^3 + a^2$. Is Clarice's work correct? Explain.